

2.1 The Cell Membrane

Every living cell is separated from its surroundings by a thin boundary that controls what enters and what leaves. This boundary is not a rigid wall but a flexible, self-sealing film that constantly rearranges itself as the cell grows, divides and responds to signals.

The membrane is built mainly from phospholipids, molecules with a water-loving head and two water-fearing tails. In water they assemble into a double layer with the tails hidden inside and the heads facing outward.

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When the balance of water on either side changes, the cell may swell or shrink. Plants, fungi and many bacteria resist bursting because a stiff outer wall surrounds the softer inner boundary.